DIGIT-PARTITIONING PRIMES AND THE ALIGNMENT FORMULA

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Abstract. For a prime p not dividing a positional base b, the fractions $\{k/p : 1 \leq k \leq p-1\}$ partition into cosets of the subgroup $\langle b \rangle \leq (\mathbb{Z}/p\mathbb{Z})^*$, each coset determining a family of repetends related by cyclic permutation. We say that p is *digit-partitioning* in base b if no two fractions from the full set $\{k/p\}$ share a digit at any common repetend position unless they are identical.

We prove that p is digit-partitioning in base b if and only if $p \leq b+1$. The proof rests on the injectivity of the digit function $r \mapsto \lfloor br/p \rfloor$ on $\{1, \ldots, p-1\}$.

For every digit-partitioning prime, the repetend alignment of $n = pm$ (with m a b-smooth integer) equals exactly $(2m-1)/(pm-1)$. This universal formula, combined with the golden threshold analysis of the companion paper, confirms that the golden ratio selects $p = 3$ uniquely among primes $p \geq 3$: the alignment limit $2/p$ exceeds $1/\varphi$ only for $p = 2$ and $p = 3$.

1. Introduction

This paper studies the digit function $\delta(r) = \lfloor br/p \rfloor$ and its structural invariants.

In a companion paper [1], we proved that the repetend alignment of $n = 3m$ (with m a 10-smooth integer) equals $(2m-1)/(3m-1)$, and that the golden ratio is the unique self-referential threshold classifying such integers. The formula arose from the observation that non-matching fractions contribute zero position-wise alignment with the base repetend.

This paper identifies the exact condition under which zero cross-alignment holds for a general prime p in a general base b. The condition is $p \leq b+1$. When this holds, the alignment formula $(2m-1)/(pm-1)$ generalizes without modification. Among primes $p \geq 3$, the alignment limit $2/p$ exceeds $1/\varphi$ only for $p = 3$.

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2. Setting

Let $b \geq 2$ be a positional base and p a prime not dividing b. The multiplicative order $L = \mathrm{ord}_p(b)$ is the length of the repetend of $1/p$ in base b [4]. The group $(\mathbb{Z}/p\mathbb{Z})^*$ has order $p-1$, and the cyclic subgroup $\langle b \rangle$ has order L, partitioning $\{1, \ldots, p-1\}$ into $(p-1)/L$ cosets. Within each coset, the repetends of k/p are cyclic permutations of a common string.

Definition 1. An integer m is *b-smooth* if every prime factor of m divides b.

Definition 2. For $n = pm$ with m a b-smooth integer, the *repetend alignment* $\alpha(n)$ is the mean, over $k \in \{1, \ldots, n-1\}$, of the position-wise digit-match proportion between the repetend of k/n and the repetend of $1/n$. Terminating fractions contribute alignment 1.

3. The Digit Function

The central object is the digit function that maps remainders to digits during long division. The floor function $\lfloor br/p \rfloor$ partitions the integers $\{1, \ldots, p-1\}$ into contiguous bins of nearly equal size [2].

Definition 3. For a prime p not dividing b, the *digit function* is

$$\delta(r) = \left\lfloor \frac{br}{p} \right\rfloor, \qquad r \in \{0, 1, \ldots, p-1\}.$$

This is the digit produced when the remainder r is processed in one step of long division of any fraction with denominator p in base b.

Lemma 4. *If $p \leq b+1$, then δ is injective on $\{1, \ldots, p-1\}$.*

Proof. We treat two cases.

Case $p = b+1$. For $r \in \{1, \ldots, b\}$,

$$\delta(r) = \left\lfloor \frac{br}{b+1} \right\rfloor = \left\lfloor r - \frac{r}{b+1} \right\rfloor = r - 1,$$

since $0 < r/(b+1) < 1$ for $1 \leq r \leq b$. The map $r \mapsto r-1$ is a bijection from $\{1, \ldots, b\}$ to $\{0, \ldots, b-1\}$.

Case $p \leq b$. For distinct $r, s \in \{1, \ldots, p-1\}$ with $r < s$, the values br/p and bs/p differ by $b(s-r)/p \geq b/p \geq b/b = 1$. If $b(s-r)/p > 1$, the floors are necessarily distinct. Equality $b(s-r)/p = 1$ requires $s - r = p/b$, hence $b \mid p$. Since p is prime and $b \geq 2$, this forces $b = p$, contradicting $\gcd(p, b) = 1$. When $b(s-r)/p$ is exactly 1, we need both br/p and $bs/p = br/p + 1$ to have the same floor, which requires br/p

to be an integer. But $\gcd(b, p) = 1$ and $1 \leq r < p$ ensure $br/p \notin \mathbb{Z}$. In all cases, $\delta(r) \neq \delta(s)$. □

Remark 5. When $p > b + 1$, the digit function maps $p - 1 > b$ distinct remainders into b digit values. By the pigeonhole principle [3], δ cannot be injective, and collisions are unavoidable.

4. The Digit-Partitioning Property

Definition 6. A prime p is *digit-partitioning* in base b if no two distinct fractions in $\{k/p : 1 \leq k \leq p-1\}$ share a digit at any common repetend position, unless their repetends are identical.

Theorem 7. *Let p be a prime not dividing b. Then p is digit-partitioning in base b if and only if $p \leq b + 1$.*

Proof. At position j of the repetend of k/p, the digit is $\delta(b^j k \bmod p)$. For distinct fractions k/p and ℓ/p with $k \not\equiv \ell \pmod p$, the arguments $b^j k$ and $b^j \ell$ are distinct modulo p (since b^j is invertible).

If $p \leq b + 1$, then δ is injective on $\{1, \ldots, p-1\}$ by Lemma 4, so distinct arguments produce distinct digits at every position. Therefore p is digit-partitioning.

If $p > b + 1$, then δ maps $p - 1 > b$ remainders into b values. By the pigeonhole principle, there exist distinct $r, s \in \{1, \ldots, p-1\}$ with $\delta(r) = \delta(s)$. The first repetend digit of r/p equals the first repetend digit of s/p, so p is not digit-partitioning. □

5. The Universal Alignment Formula

Proposition 8. *For $n = pm$ with m a b-smooth integer and p not dividing b, the non-terminating fractions in $\{k/n : 1 \leq k \leq n-1\}$ produce exactly $p - 1$ distinct repetends.*

Proof. The $m - 1$ multiples of p in $\{1, \ldots, pm-1\}$ give terminating fractions. The remaining $(p-1)m$ values of k are not divisible by p.

Choose t with $m \mid b^t$, and write $b^t = mu$ where $\gcd(u, p) = 1$ (possible since m is b-smooth and $p \nmid b$). Then

$$\frac{k}{pm} = \frac{ku}{p \cdot b^t},$$

so after t base-b digits, the repetend of $k/(pm)$ is exactly the repetend of $(ku \bmod p)/p$. Since multiplication by u permutes $(\mathbb{Z}/p\mathbb{Z})^*$, the non-terminating repetends of $\{k/(pm)\}$ are in bijection with the repetends of $\{r/p : 1 \leq r \leq p-1\}$.

For digit-partitioning primes ($p \leq b+1$), the injectivity of δ (Lemma 4) ensures that distinct remainders produce distinct digits at every position,

so distinct starting remainders yield distinct repetends. There are therefore exactly $p-1$ distinct repetends.

For general primes $p > b+1$, the number of distinct repetends equals $(p-1)/L$ where $L = \mathrm{ord}_p(b)$, since each coset of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$ produces one repetend. When b is a primitive root modulo p (so $L = p-1$), the count is again $p-1$.

Each residue class modulo p contains exactly m elements of $\{1, \ldots, pm-1\} \setminus p\mathbb{Z}$, so each repetend is shared by exactly m fractions. □

Theorem 9. *If $p \leq b+1$ and $m \geq 2$ is b-smooth, then*

$$\alpha(pm) = \frac{2m-1}{pm-1}.$$

Proof. By Lemma 8, the $pm-1$ fractions partition into:

(A) $m-1$ terminating fractions (multiples of p). Each contributes alignment 1.
(B) m fractions whose cycle matches $1/(pm)$ (one of the $p-1$ cycle classes). Each contributes alignment 1.
(C) $(p-2)m$ fractions in the remaining $p-2$ cycle classes. Since p is digit-partitioning, distinct cycles share no digit at any common position (Theorem 7). Each contributes alignment 0.

Total: $(m-1) + m = 2m-1$ aligned out of $pm-1$. □

Corollary 10. *For fixed p, the alignment $\alpha(pm) \to 2/p$ as $m \to \infty$.*

6. Enumeration

For each base b, the digit-partitioning primes are precisely the primes $p \leq b+1$ that do not divide b.

Base b	Digit-part. primes ($p \leq b{+}1$, $p \nmid b$)	Alignment limit $2/p$
4	$3, 5$	0.667, 0.400
6	$5, 7$	0.400, 0.286
8	$3, 5, 7$	0.667, 0.400, 0.286
10	$3, 7, 11$	0.667, 0.286, 0.182
12	$5, 7, 11, 13$	0.400, 0.286, 0.182, 0.154
16	$3, 5, 7, 11, 13, 17$	0.667, ..., 0.118

Table 1. Digit-partitioning primes for selected bases and their alignment limits $2/p$.

In base 10, the digit-partitioning primes are 3, 7, and 11. These correspond to the three classical mechanisms identified in earlier analysis:

- $p = 3$: $\mathrm{ord}_3(10) = 1$ (single-digit repetends).
- $p = 7$: $\mathrm{ord}_7(10) = 6 = p-1$ (cyclic number, primitive root).
- $p = 11$: $\mathrm{ord}_{11}(10) = 2$, with $10 \equiv -1 \pmod{11}$ (nines-complement partition).

These mechanisms are unified by the single condition $p \leq b + 1$.

7. Connection to the Golden Ratio

The alignment limit $2/p$ exceeds $1/\varphi$ only when $p < 2\varphi \approx 3.236$. Among primes $p \geq 3$, only $p = 3$ meets this condition. (For $p = 2$, the alignment limit is 1, trivially above any threshold.)

The self-referential characterization of [1] applies: the golden ratio's minimal polynomial $\tau^2 + \tau - 1$ divides the self-referential cubic if and only if $p = 3$, producing the fourth-power identity $\tau^4 = 2 - 3\tau$.

The present paper strengthens this result. The formula $(2m-1)/(pm-1)$ is not a special case tied to base 10 or to any particular mechanism. It holds universally for all digit-partitioning primes in all bases, and the characterization $p \leq b + 1$ shows that the class of such primes is determined by a single inequality. Among primes $p \geq 3$ in this class, $p = 3$ is the unique value with alignment limit above $1/\varphi$.

8. Remarks

Structural manifestations. Although the characterization is a single inequality, the digit-partitioning property manifests through different structural mechanisms depending on the relationship between b and p:

- When $b \equiv 1 \pmod{p}$: repetends have length one, and each fraction uses a unique single digit.
- When $p = b+1$ (so $b \equiv -1 \pmod{p}$): repetends have length two, and cosets correspond to $(b-1)$-complement pairs, partitioning the digit set completely.
- When b is a primitive root modulo p and $p \leq b+1$: there is a single coset, and the repetend is a cyclic number with all distinct digits.

These are consequences of the injectivity of δ, not independent conditions.

The boundary case $p = b + 1$. The case $p = b + 1$ is the tightest: δ maps b remainders into b digit values, and injectivity holds by a narrow margin. In base 10, this gives $p = 11$, whose five nines-complement pairs partition all ten digits exactly. The boundary case always produces the complement-pair structure, since $b + 1 = p$ implies $b \equiv -1 \pmod{p}$.

Open questions. For $p > b + 1$, the alignment formula acquires correction terms from digit collisions. The corrections are governed by the squared bin sizes of the digit function and have a closed form for primitive-root primes.

References

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